

Project Documentation: Bank Finance Management System

1. Project Overview

The **Bank Finance Management System** is a Django-based web application designed to manage banking operations, including user accounts, transaction management, and financial predictions using machine learning. The system enables both administrators and customers to perform various banking activities securely and efficiently.

2. Technology Stack

- **Frontend:** HTML, CSS, JavaScript (templates rendered by Django)
- **Backend:** Python, Django Framework
- **Database:** SQLite (default Django DB)
- **Machine Learning:** Scikit-learn (in `predict.py`)

3. Project Structure

```
bank_project/
├── bank_project/      # Django settings and URLs
├── finance/           # Core app for banking features
│   ├── admin.py       # Admin site definitions
│   ├── forms.py       # Django forms
│   ├── models.py      # Database models
│   ├── predict.py     # ML prediction logic
│   ├── views.py       # View logic and controllers
│   ├── urls.py        # URL routing for the app
│   └── templatetags/  # Custom Django template filters
└── manage.py         # Django management script
```

4. Key Features

- User authentication and roles (admin, user)
- Transaction history management
- Bank account details and balance inquiry
- Machine learning prediction (e.g., fraud detection, credit scoring)
- Admin dashboard for monitoring

5. Setup Instructions

```
# Step 1: Clone the project
$ git clone <repository_url>
```

Step 2: Navigate to the project

```
$ cd bank_project
```

Step 3: Create a virtual environment

```
$ python -m venv env
```

```
$ source env/bin/activate # On Windows: env\Scripts\activate
```

Step 4: Install dependencies

```
$ pip install -r requirements.txt
```

Step 5: Apply migrations

```
$ python manage.py migrate
```

Step 6: Run the server

```
$ python manage.py runserver
```

6. Usage Guide

- Access the application via <http://127.0.0.1:8000/>
- Register or log in using provided credentials
- Users can view balance

7. ML Component

The `predict.py` file uses a trained model (e.g., logistic regression or decision tree) to make predictions based on user input. This could be used for:

- Fraud detection
- Loan approval chances
- Credit scoring

8. Contributing

Contributions are welcome! Please follow the typical GitHub workflow:

- Fork the repo
- Create a new branch
- Submit a pull request with detailed description

9. License

This project is licensed under the MIT License.

10. Credits

Developed by [SYNTAX SQUAD].